Assignment No.4 2413FEB2T1 : Engineering Mathematics-2
Topic: Numerical Integration

- Trapezoidal Rule
  $I = \int_a^b y dx = \frac{h}{2}[X + 2R]$
- Simpson's $(\frac{1}{3})^{rd}$ Rule
  $I = \int_a^b y dx = \frac{h}{3}[X + 4O + 2E]$
- Simpson's $(\frac{3}{8})^{th}$ Rule
  $I = \int_a^b y dx = \frac{3h}{8}[X + 2T + 3R]$

Solve the following:

1. A rocket is launched from the ground. Its acceleration is registered during the first 80 seconds and is given in the following table.

t(sec)	0	10	20	30	40	50	60	70	80
a(m/s^2)	30	31.63	33.34	35.47	37.75	40.33	43.25	46.69	50.67

   Find the velocity of rocket at t = 80 sec by
   (a) Trapezoidal Rule and
   (b) Simpson's $(\frac{1}{3})^{rd}$ Rule
2. Evaluate $\int_0^2 e^x dx$by Trapezoidal rule with 5 subintervals.
3. Evaluate $\int_{0.2}^{1.4} (sinx - log_e x + e^x) dx$ using
   (a) Trapezoidal rule,
   (b) Simpson's $1^{rd}/3$ rule, and
   (c) Simpson's $3^{th}/8$ rule.
4. Find the value of $\int_0^1 \frac{x^2}{1+x^3} dx$ using
   (a) Trapezoidal Rule and
   (b) Simpson's $(\frac{1}{3})^{rd}$ Rule. Compare the Errors with exact value of integral.
5. Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using
   (a) Trapezoidal rule,
   (b) Simpson's $1^{rd}/3$ rule, and
   (c) Simpson's $3^{th}/8$ rule.
6. Using the following data, evaluate $\int_1^7 y dx$ by trapezoidal rule.

x	1	2	3	4	5	6	7
y	2.105	2.808	3.614	4.604	5.857	7.451	9.467

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7. Evaluate $\int_0^{\pi} \sin x \, dx$ by Trapezoidal rule is 10 subintervals.

8. Evaluate $\int_1^7 y \, dx$ by

   (a) Trapezoidal rule

   (b) Simpson's $1^{rd}/3$ rule

   (c) Simpson's $3^{th}/8$ rule.

   Where y is given as

x	1	2	3	4	5	6	7
y	2.157	3.519	4.198	4.539	4.708	4.792	4.835

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